

Question ID: 1f0966db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The function f is defined by $f(x) = \frac{9}{7}x + \frac{8}{7}$. For what value of x does $f(x) = 5$?

Correct Answer: 3

Rationale

The correct answer is **3**. Substituting **5** for $f(x)$ in the given function yields $5 = \frac{9}{7}x + \frac{8}{7}$. Multiplying each side of this equation by **7** yields $7(5) = 7(\frac{9}{7}x + \frac{8}{7})$, or $35 = 9x + 8$. Subtracting **8** from each side of this equation yields $27 = 9x$. Dividing each side of this equation by **9** yields $3 = x$. Therefore, $f(x) = 5$ when the value of x is **3**.